HAT-P-12b

R-1272

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-12_210209` · created 2026-07-31T14:00:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459254.8502 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.014
Transit depth (Rp/Rs)^2	1.84 +/- 0.37 [%]
Orbital Inclination (inc)	89.93 +/- 0.75
Airmass coefficient 1 (a1)	1.162 +/- 0.018
Airmass coefficient 2 (a2)	-0.1174 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	None
Optimal Aperture	3.88
Optimal Annulus	5.64
Transit Duration (day)	0.0979 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.136 ± 0.014 vs 0.1406 (deviation 0.28σ)
Duration fit vs published	0.0979 d vs 0.0975 d (deviation 0.19σ)
Mid-time O–C	2.7 min
Depth SNR	8.3
Residual scatter	1.51 %

Light curve

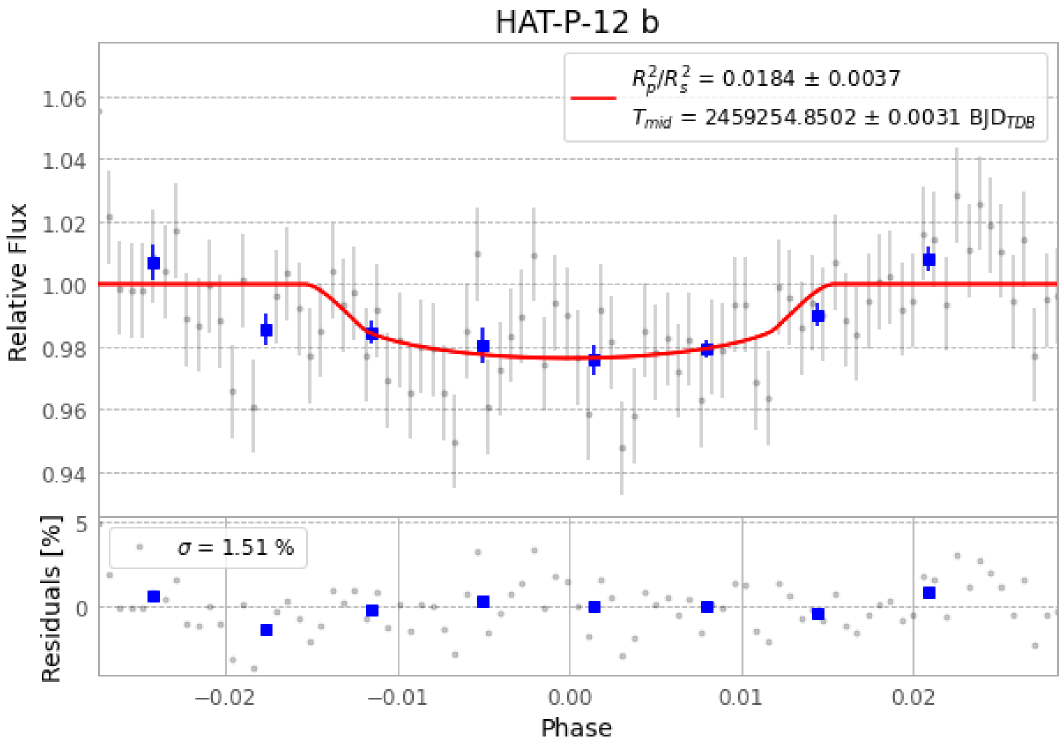

Figure 1 — FinalLightCurve_HAT-P-12 b_2021-02-08.png (SHA-256 2ecb93fd1dc4220b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 243], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d7f604821412e8bb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

87 FITS frames (57 MB), `HATP-12210209061212.FITS` → `HATP-12210209103012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210209.log (51c716135aca…), FinalLightCurve_HAT-P-12 b_2021-02-08.png (2ecb93fd1dc4…), FinalLightCurve_HAT-P-12 b_2021-02-08.csv (d6ee527e0c44…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:00Z.